

Session 2: How can I import data?

A Slower Introduction to R

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Scripts

Yet another cooking analogy

- sometimes I can't quite remember what I did in the kitchen
- what is a possible solution?

Yet another cooking analogy

- using the console is like winging it in the kitchen

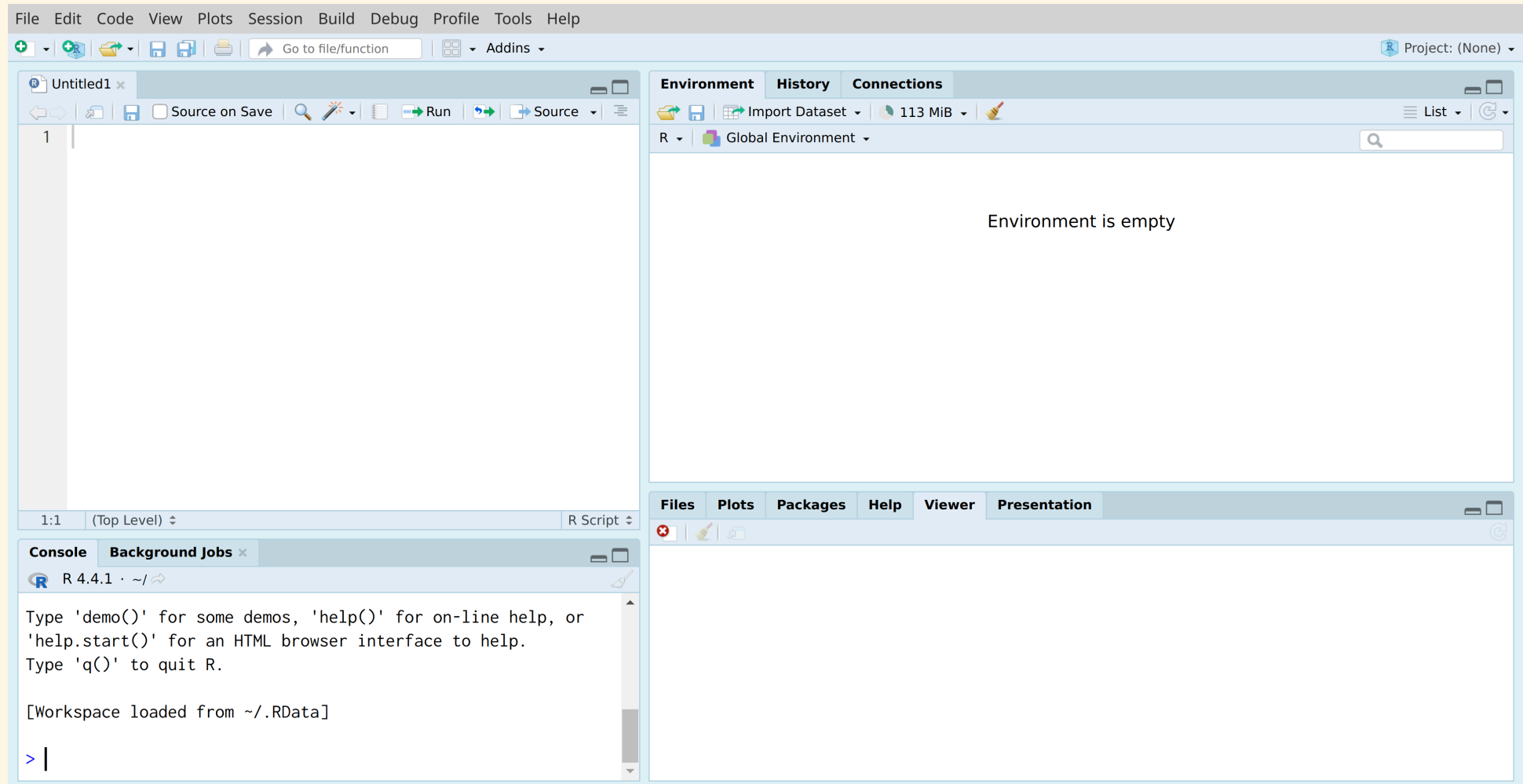
What are scripts?

- scripts = documents that you can use to store code
- continuing the analogy, scripts are like recipes

Why should I use scripts?

- scripts are a way for you to save and revisit your work, much the same way a recipe would allow you to cook the same dish again
- scripts help promote transparency and reproducibility

Scripts in RStudio



Running scripts

- you can **send code** from scripts to your computer
- this process is known as **running** the code, and will result in the code and output showing up in the console

Live demonstration



Running code

1. **Select** the code.
2. **Run** the code, using either the Run icon at the top of the pane or the keyboard shortcut (**ctrl + enter** on Windows or **cmd + return** on Mac).

One line for each statement



Each statement (“sentence”) of code should be on its own line.

- for example, you cannot do this:

```
sqrt(81) pi^2
```

- you should, instead, do this:

```
sqrt(81)  
pi^2
```

Long statements

- if you wish, you can write a statement of code that spans more than one line:

```
(0.25*0.4)/  
(0.25*0.4+(1-0.9)*(1-0.4))
```

Long statements

Caution

If you are writing a sentence of code that spans more than one line, always **place mathematical operators before the line break** rather than after the line break.

Long statements

- this will work as intended:

```
5*12+  
2
```

- this **will not work as intended** and will not result in an error:

```
5*12  
+2
```

- **why?**

Comments

- you can include human-language comments in scripts
- comments serve as notes for yourself and for your collaborators

Adding comments

- to include comments in scripts, begin the line with the pound sign

```
# calculate positive predictive value when prevalence is 0.4  
(0.25*0.4)/(0.25*0.4+(1-0.9)*(1-0.4))  
  
# calculate positive predictive value when prevalence is 0.3  
(0.25*0.3)/(0.25*0.3+(1-0.9)*(1-0.3))
```


Live demonstration



Saving scripts

1. Click the **save icon** at the top of the pane or, equivalently, select **File > Save**.
2. Select a folder if you don't want to save the file to the folder being suggested to you.
3. Pick a filename, making sure to add **.r** or **.R** to the end.

Exercise 1

Exercise 1

1. Check whether or not a blank script is already open (the tab might be labeled Untitled1). If not, open a new script by selecting File > New File > R Script.
2. A common cutoff for a borderline high LDL cholesterol level is 130 mg/dL. In the first line of your script, type code that stores the value of 130 as `ldl_borderline`, using object assignment.

Exercise 1

3. A common cutoff for a high LDL cholesterol level is 160 mg/dL. In the second line, type code that stores the value of 160 as `ldl_high`, using object assignment.
4. Add a comment above each object assignment that describes the object assignment. For readability, include a blank line between the first object assignment and the second comment.

Exercise 1

5. Select and run your code.
6. In the console, check to see that `ld1_borderline` and `ld1_high` were properly defined (type each object name and press the enter or return key on your keyboard).
7. Save the script, picking some filename that makes sense to you (such as `session_2.r`). Make sure to add `.r` or `.R` at the end. You might first create a folder on your computer for this workshop series.

Suggested workflow

Slide formatting

- to facilitate copying-and-pasting, throughout the remainder of the workshop series, I will omit the prompt
- for example:

```
(0.25*0.4)/(0.25*0.4+(1-0.9)*(1-0.4))  
#| [1] 0.625
```

- this does not necessarily mean that you should use a script over a console
- what do I suggest?

Should I use the script or the console?



Go **back and forth** between the script and the console.

- there are two ways to do this:
 1. try things out in the console and then copy and paste code into your script
 2. type in the script, run your code, and examine the output in the console

Should I use the script or the console?

- as you get more experienced with R, you will probably do most of your typing in scripts

How can I save my work?



Tip

Use **scripts** as a way to save work!

- I discourage you from trying to save work by relying on objects
- instead, **lean on scripts**

How should I structure my scripts?



Your scripts should have a **sequence**.

- as with the recipes analogy, the idea is that the script should serve as a **record of your work**, and have a general **top-to-bottom flow**

How should I keep my scripts clean?



Only include **code and comments**.

- do not include un-commented human-language text!
- do not include output!

How can I facilitate readability?



Tip

Write comments.

How can I facilitate readability?

- I like to write short but frequent comments
- I like to begin every comment with a verb

```
# import data
nhanes<-read.csv('~Downloads/data/acs/2019/psam_p06.csv.gz')

# rename variables
names(nhanes)<-tolower(names(nhanes))

# subset to individuals 18-65 years of age
nhanes<-subset(nhanes,agep>=18&agep<=65)
```

How can I facilitate readability?



"I kinda thought the subtitles would be more helpful."

What if my script is really long?



Consider using more than one script.

- the scripts should have a natural sequence from one to another, such as:
 1. `import and prepare data.r`
 2. `analyze data.r`

How can I write good code and avoid mistakes?



Tip

Continually **revisit** your scripts!

- think of a script as being **like a work-in-progress manuscript**
- be willing to **continually edit it**

Importing data

NHANES

- the National Health and Nutrition Examination Survey (NHANES) is a large, CDC survey of American health:

The National Health and Nutrition Examination Survey (NHANES) collects data about the health of adults and children in the United States. We also collect data about what participants eat, drink, and take as supplements to determine how many nutrients are in their diet.

NHANES

- [nhanes_1.csv](#), available on the CLE, contains data from the 2021–2023 implementation of NHANES
- I created the file by combining ([merging](#)) several CDC data files, from across several dimensions
- I restricted the data to [individuals 18 years of age or older](#)
- I added a handful of variables, using existing variables
- the original data, and [data descriptions](#), are available [here](#)

Importing the data

- as mentioned in the orientation, there are often many ways to do something in R
- I'm going to show you how to use RStudio's drop-down menu to import the data
- as a preview, this process will result in object assignment

Live demonstration



Importing the data

- in our quick demonstration we changed two parts of the pop-up window:
 - **Name**: the name from the resulting object assignment (using the jar analogy, this is the label you put on the jar)
 - **Heading**: an option that indicates whether or not variables are included as the first line of the data file

The corresponding code

- the import tool will print the corresponding R code in the console
- that code involves object assignment

The corresponding code



Tip

Examine the code to begin to understand it.

The corresponding code



Copy and paste the corresponding code to a script so you can later reproduce the import without having to rely on the import tool.

- it is not necessary to copy the `View()` portion of the code

The view tab

- by default, the import tool will automatically show the data in a new tab in the Source pane
- you can **close the tab** if you want (the object itself will not be removed)
- you can always open the view tab again by using **View()**

The view tab



Examine the data carefully after importing a data file for the first time.

- check that **variable names** are there (if they should be there)
- scroll through to check to see that columns are lined up properly

Other ways to explore the data

- `names()`: for all the variable names
- `head()`: for the first few rows
- `tail()`: for the last few rows
- `str()`: for a glimpse of the data that shows variable names, object classes (more about this later), and the first few values

Other ways to explore the data

- `dim()`: for the number of rows and the number of columns, in that order
- `nrow()`: for the number of rows
- `ncol()`: for the number of columns

Other ways to explore the data

```
dim(nhanes)
#| [1] 8153 195
nrow(nhanes)
#| [1] 8153
ncol(nhanes)
#| [1] 195
```


Exercise 2

Exercise 2

1. Download the data from the CLE, by clicking on the link, and then browsing for a folder on your computer where you would like to save the file. If the data opens in your browser instead, go back to the CLE page and instead right-click on the link (or control-click or two-finger click), and select save.
2. Use RStudio's drop-down menu to import the data: File > Import Dataset > From Text (base). Please make sure to select the base option.

Exercise 2

3. In the pop-up window, select Yes under the Heading option.
4. Confirm that the data look OK in the preview section at the bottom right. If so, select the Import button.

Exercise 2

5. Confirm that the data look OK in the viewer that pops up (in the Source pane). You can optionally close the tab when you're done.
6. Copy and paste the code for importing the data into your script.
7. Use `head()` to view the first few rows of the data.

Dollar-sign notation

An error message

- the `mean()` function will find the mean of a set of numbers
- there is a variable in the data called `ridageyr`, which stores ages in years
- we might be inclined to try this:

```
mean(ridageyr)
#| Error:
#| ! object 'ridageyr' not found
```

- **why** do we get an error message?

An error message

- we get an error message because R allows multiple data sets to be available at a time
- so, any time we refer to a variable we also have to refer to the data set

Dollar-sign notation

```
mean(nhanes$ridageyr)
#| [1] 52.14436
```

- `mean()`: a function for finding the arithmetic mean
- `nhanes`: the label we gave for our data when we imported it
- `$`: indicates that `ridageyr` is a column in `nhanes`
- `ridageyr`: the variable name

Exercise 3

Exercise 3

1. The `head()` and `tail()` functions, mentioned earlier, can be used not only to preview data but also to preview variables within data. Use these functions to view the first few, and the last few, values of the `gender` variable.
2. The `range()` function will show the range of a variable (this is, the minimum and maximum values). Use `range()` to find the age range of individuals in the NHANES data (the age variable is `ridageyr`).

Object classes

Another error message

```
mean(nhanes$gender)
#| Warning in mean.default(nhanes$gender): argument is not numeric or logical:
#| returning NA
#| [1] NA
```

- there is a variable in the NHANES data called `gender`
- we are using dollar-sign notation
- so, *why* does this result in an error?

Another error message

- we got an error message because gender does not include numbers:

```
head(nhanes$gender)
#| [1] "Male"  "Male"  "Female" "Male"  "Female" "Male"
```

Object classes

- a more general way to check is to use the `class()` function:

```
class(nhanes$gender)
#| [1] "character"
```

Object classes

- the `class()` function reports the object type
- R has many different object types
- the formal term for object types is object classes
- using the jar analogy, you can think of object classes as being like the different types of ingredients you might have inside of jars

A taxonomy of object classes



Vectors

```
class(nhanes$ridageyr)
#| [1] "integer"
is.vector(nhanes$ridageyr)
#| [1] TRUE
class(nhanes$gender)
#| [1] "character"
is.vector(nhanes$gender)
#| [1] TRUE
```

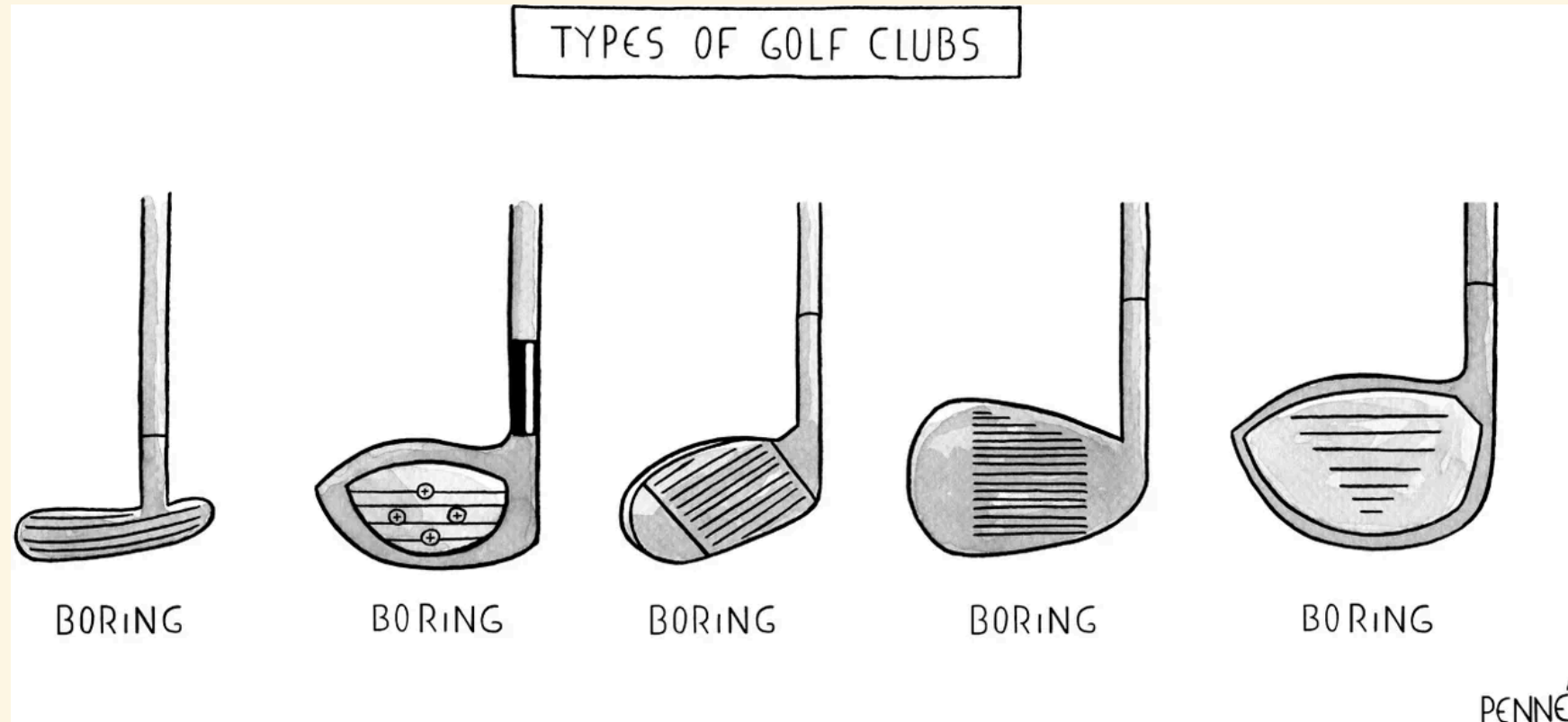
- **integer** and **character** objects are examples of **vectors**

Numeric objects

```
is.numeric(nhanes$ridageyr)
#| [1] TRUE
is.integer(nhanes$ridageyr)
#| [1] TRUE
is.numeric(22.52)
#| [1] TRUE
is.integer(22.52)
#| [1] FALSE
```

- **integer** objects are themselves examples of **numeric** objects, but not all numeric objects are integers

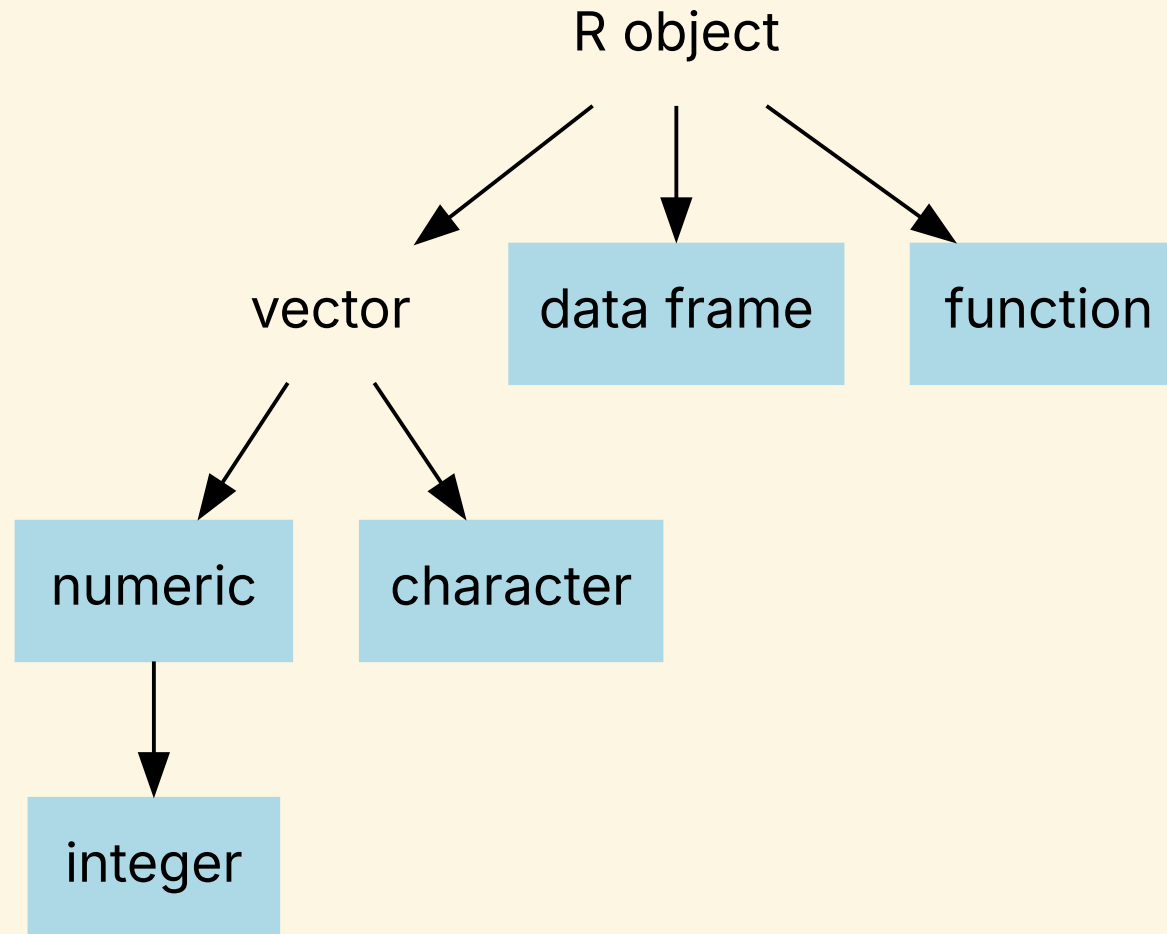
A taxonomy of object classes



Data frames and functions

```
class(nhanes)
#| [1] "data.frame"
class(mean)
#| [1] "function"
```

A taxonomy of object classes



Object classes

- mostly I just want you to begin to be aware of all of this because **classes will determine what you can and cannot do** with objects, as illustrated in the first example

Exercise 4

Exercise 4

1. The `bpxodi1` variable stores measurements of diastolic blood pressure. Try to find the mean of this variable. You should see a value of `NA`.
2. Now check the class of this variable. You should see that the variable is in fact a numeric object.
3. Finally, use `head()` to take a look at the first few values of the variable. Why do you think might `mean()` returned `NA` in the first problem here? We'll talk next week about how to approach this!